



ReactJS



JSX

JSX is a syntax extension for JavaScript that lets you write HTML-like markup inside a JavaScript file

Rules

Component

Components are one of the core concepts of React. They are the foundation upon which you build user interfaces (UI).

A React component is a JavaScript function that you can sprinkle with markup

Multiple Components

Curly Braces

JSX lets you write HTML-like markup inside a JavaScript file, keeping rendering logic and content in the same place. Sometimes you will want to add a little JavaScript logic or reference a dynamic property inside that markup. In this situation, you can use curly braces in your JSX to open a window to JavaScript.

Props

React components use props to communicate with each other. Every parent component can pass some information to its child components by giving them props

Props Destructing

```
function Avatar({ name, img }) {  
  // Get the complete jsx passed by the parent component  
}
```

Prop “children”

```
function Avatar({ children }) {  
  // Get the complete jsx passed by the parent component  
}
```


Conditional rendering

Your components will often need to display different things depending on different conditions. In React, you can conditionally render JSX using JavaScript syntax like if statements, `&&`, and `? :` operators

Rendering lists

Display multiple similar components from a collection of data.

Use the JavaScript array methods to manipulate an array of data.

- map
- filter

Adding Interactivity

Update things on the screen in response to user input actions like, clicking on button, enter input on textbox, checking the checkbox etc..

Responding to events

React lets you add event handlers to your JSX. Event handlers are your own functions that will be triggered in response to user interactions like clicking, hovering, focusing on form inputs, and so on

Managing State

Thinking about UI declaratively

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- Identify your component's different visual states
- Determine what triggers those state changes
- Represent the state in memory using **useState**
- Remove any non-essential state variables
- Connect the event handlers to set the state

Sharing state between components

Sometimes, you want the state of two components to always change together. To do it, remove state from both of them, move it to their closest common parent, and then pass it down to them via props. This is known as lifting state up, and it's one of the most common things you will do writing React code

MongoDB

MongoDB is a popular, source-available NoSQL database program that uses a document-oriented data model. Instead of using the rows and columns found in traditional relational databases, MongoDB stores data in flexible, JSON-like documents called BSON (Binary JSON)

Key Features

- Document Model: Maps directly to objects in your application code, making it intuitive for developers to work with.
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- Flexible Schema: Documents in the same collection do not need to have the same set of fields or structure, allowing for rapid iteration.
- Horizontal Scalability: Uses sharding to distribute data across many servers, handling massive amounts of data and high throughput.

Express JS

Express.js (or simply Express) is the most popular, minimalist, and unopinionated web framework for Node.js. It acts as a thin layer of fundamental web application features—such as routing and middleware—without obscuring the core features of Node.js

Express JS

Basics

Route handlers

Request path params, querystring and body

Response status code and json

Creating Express App

CRUD routes for movies API

Mogos

Creating Schema

Mongo APIs for CRUD operations

Middleware

Express middleware are functions that execute during the request-response cycle of an application

Example: Log requests urls and invoke next handler

Webpack Bundling

Express middleware are functions that execute during the request-response cycle of an application

Example: Log requests urls and invoke next handler